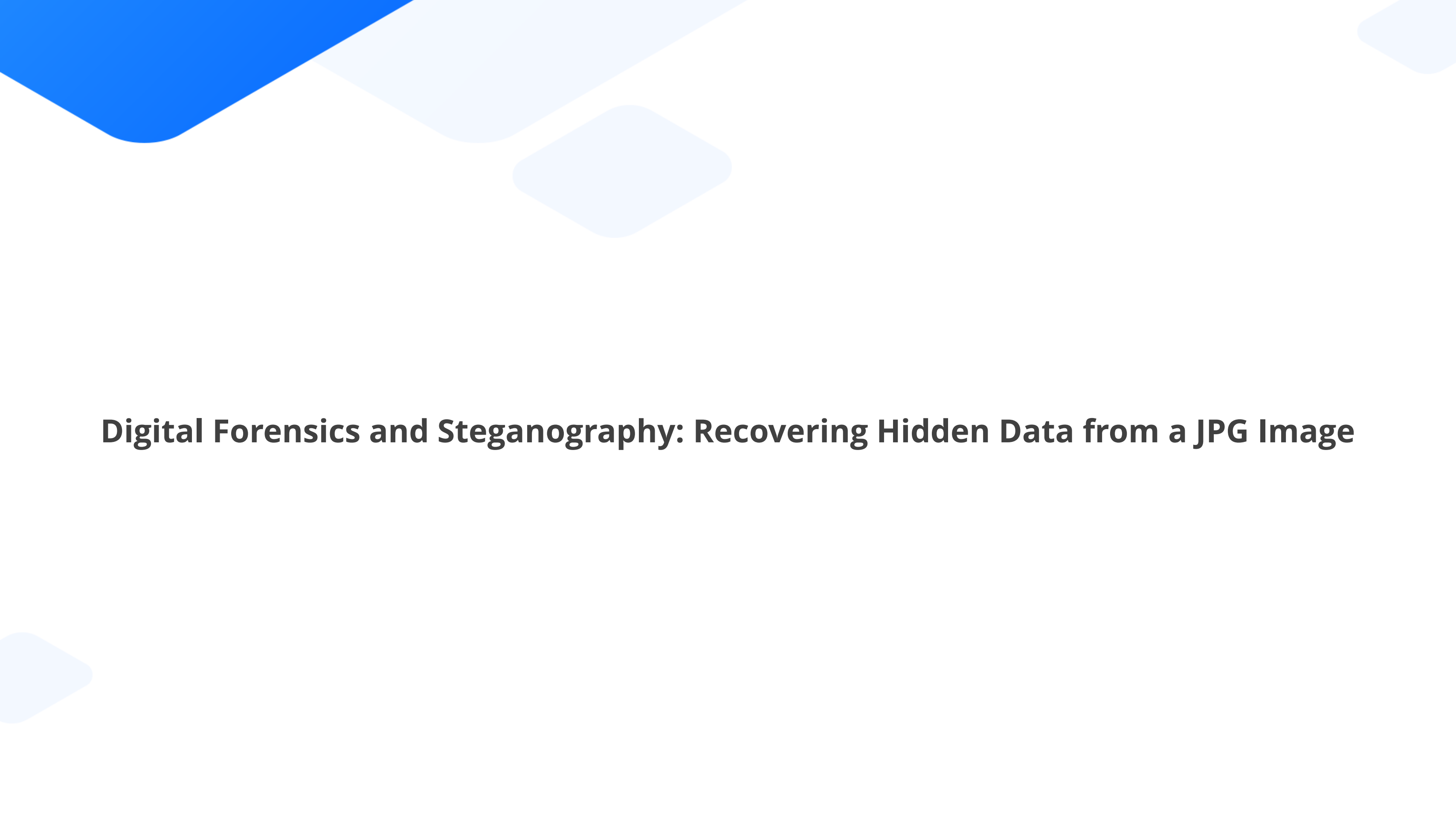


Ransomware and Malware: Defense, Analysis, and Response



Course-End Project





Digital Forensics and Steganography: Recovering Hidden Data from a JPG Image

Objective

To demonstrate the process of embedding text into a JPG image using steganography techniques and storing it on a 2-disk system to simulate real-world data storage

Description:

Simulate a scenario by deleting the file and making it inaccessible. Utilize FTK (Forensic Toolkit) to recover the deleted file from the disk image and extract the hidden text from the recovered JPG. This project highlights data-hiding methods and the forensic recovery process.



Problem Statement and Motivation



Real-time scenario:

At a leading financial institution, concerns of a potential data breach arise when suspicious JPG images exchanged in routine emails mysteriously disappear from the system. Their seemingly harmless nature makes detection challenging, but unusual patterns in network activity raise red flags for the cybersecurity team.

During an investigation, the security lead identifies a peculiar anomaly in the network logs: a deleted file with a lingering digital footprint. The team retrieves fragments of the file from deep storage and reconstructs it, revealing what appears to be an ordinary image. However, its presence raises further suspicion.

After relentless efforts, the team discovers the truth—the image contains hidden data capable of compromising the institution's security. This finding intensifies their investigation as they work to uncover the perpetrators and mitigate the risk before it escalates.

Industry Relevance

The following tools used in this project serve specific purposes within the industry:

1. **Steganography Tool (for example, SilentEye, StegHide):** Is used to embed hidden data within JPG images to demonstrate data-hiding techniques
2. **FTK (Forensic Toolkit):** Enables recovery of deleted files and analysis of disk images for digital forensic investigations
3. **File System (2GB NTFS Partition):** Simulates real-world storage environments for practicing file recovery techniques
4. **JPG Image Format:** Represents a common file format used for embedding and concealing hidden data



Industry Relevance

The following tools used in this project serve specific purposes within the industry:

5. **Disk Image (2-Disk System):** Provides a realistic scenario for recovering deleted files from storage environments



Tasks

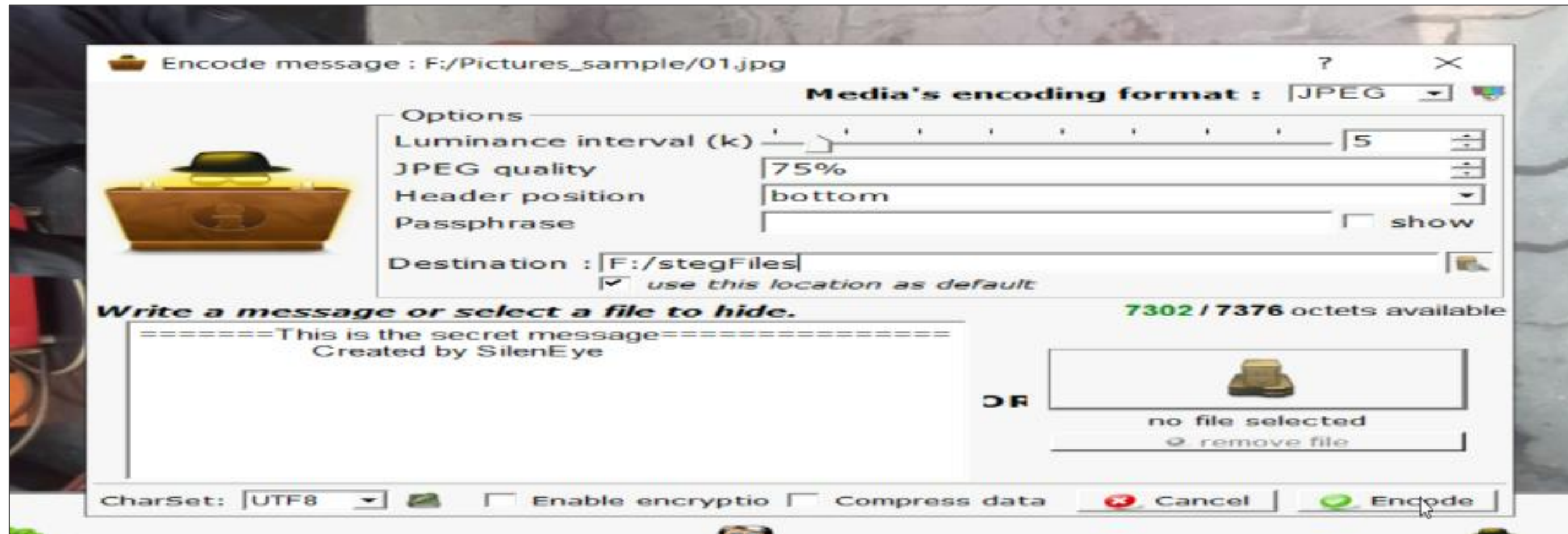
The following tasks outline the process of deploying a vulnerable web application, performing reconnaissance, and assessing its security:

1. Create a standard JPG image and embed hidden text using a steganography tool
2. Download and prepare the tools and environment for testing
3. Download and install FTK Imager
4. Place the JPG image with hidden text onto a 2GB NTFS partition
5. Delete the JPG image to simulate file removal
6. Recover the deleted image and extract hidden text



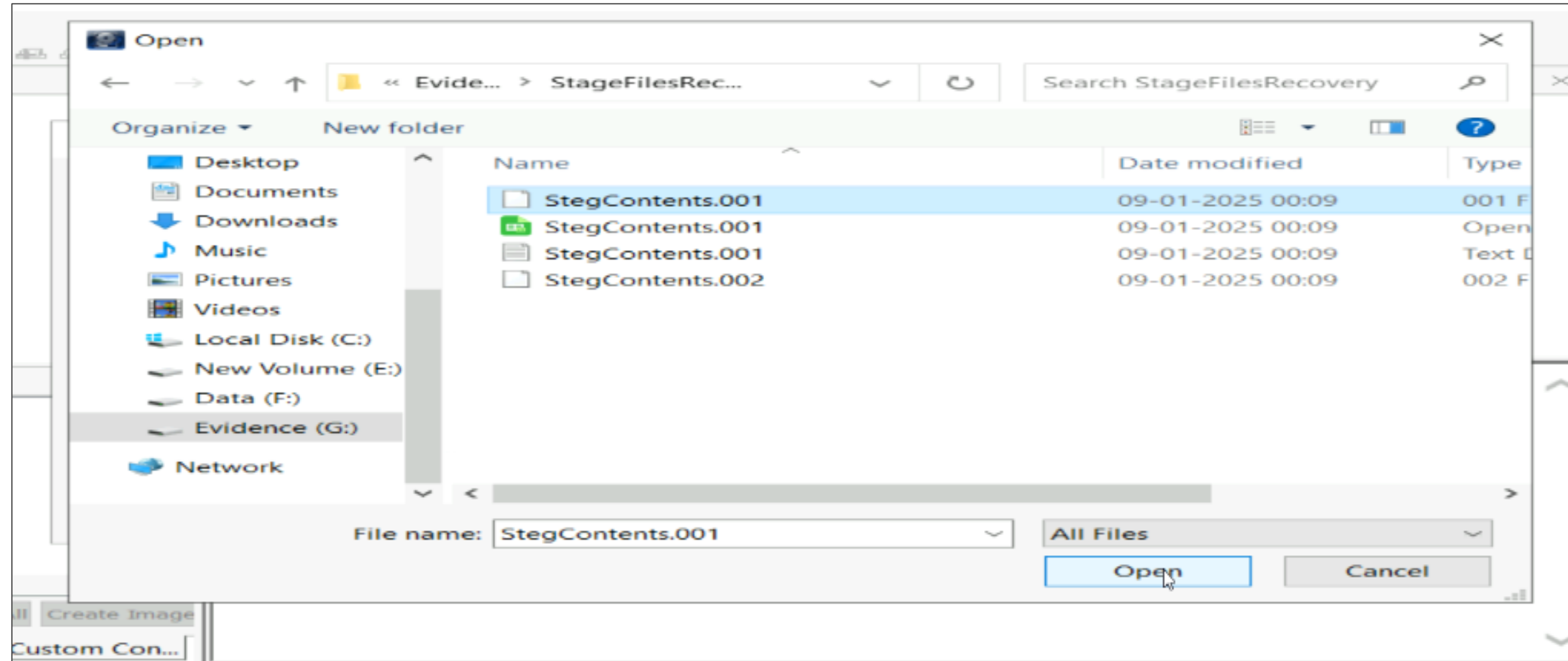
Output Screenshots

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Output Screenshots

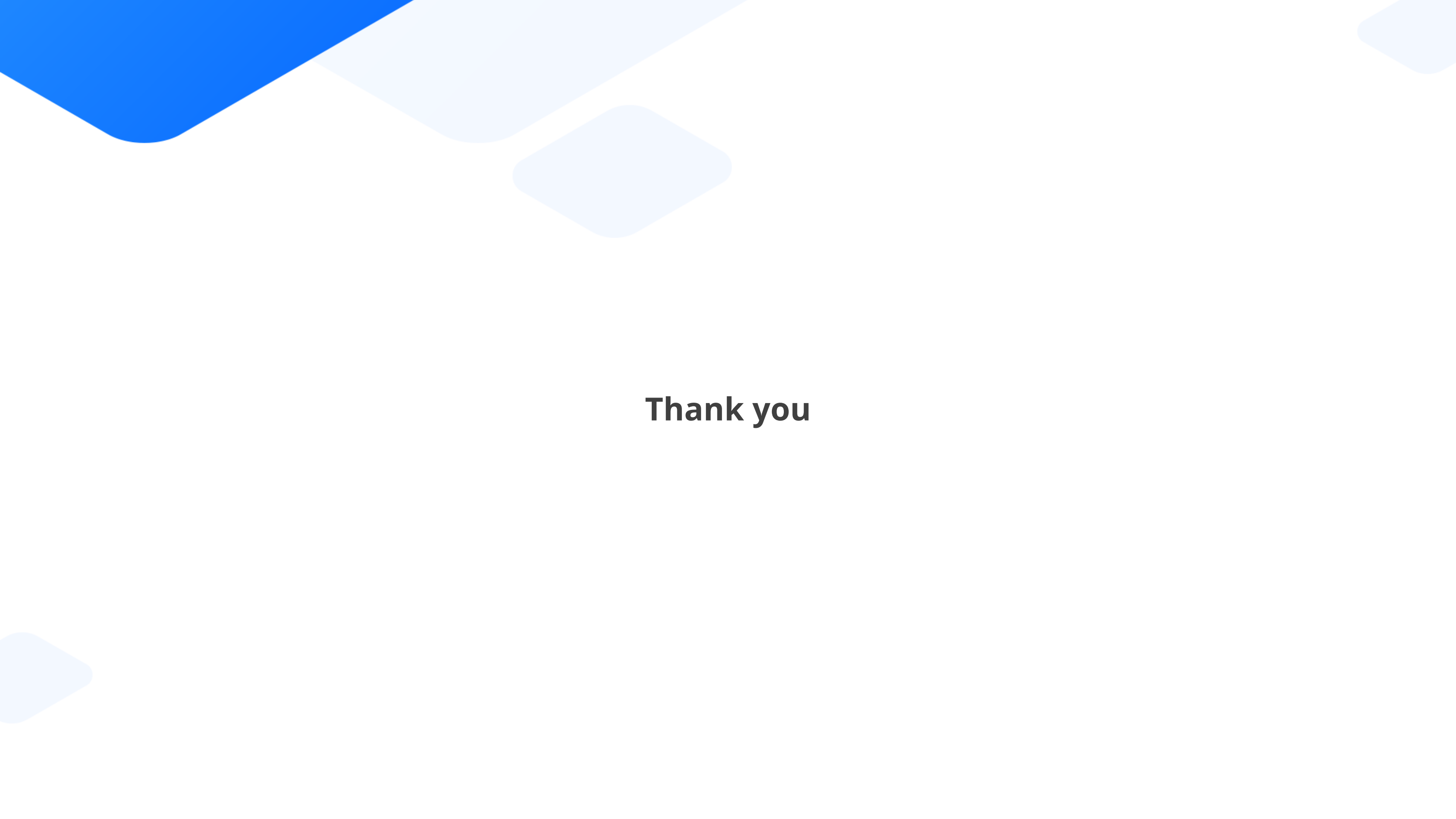
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Project References

- **Task 1:** Lesson 01 and Lesson 03
- **Task 2:** Lesson 04
- **Task 3:** Lesson 04
- **Task 4:** Lesson 04
- **Task 5:** Lesson 04
- **Task 6:** Lesson 03 and Lesson 04





Thank you